

Method Swap Challenge Teacher Notes

TOK Cognitive Science Activity Bank • Cross-AOK • 35-45 min

Category	Cross-AOK	Time	35-45 min
Difficulty	Medium	ID	method-swap-challenge

Big idea: Methods carry assumptions about what can be known and how it should be justified.

Overview

Students apply one AOK's method to another AOK's question and evaluate what breaks, transfers, or improves.

Student Prompt

Inspect Method Swap Challenge Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Method Swap Challenge Stimulus A	Student-facing stimulus 1: Question: Is this policy fair? Method card: mathematical proof. Name the AOK, the method being used, and the standard of evidence you would apply.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Method Swap Challenge Stimulus B	Student-facing stimulus 2: Question: What caused this revolution? Method card: controlled experiment. Name the AOK, the method being used, and the standard of evidence you would apply.	Use this for comparison, a second round, or a partner challenge.
Method Swap Challenge Reveal / Twist	Student-facing stimulus 3: Question: What caused this revolution? Method card: controlled experiment. Name the AOK, the method being used, and the standard of evidence you would apply.	Teacher reveal: connect the changed response to the big idea — Methods carry assumptions about what can be known and how it should be justified.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting Method Swap Challenge Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Method Swap Challenge Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.

#	Question for students	Teacher key / cue
6	How do methods shape what can be known?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how methods carry assumptions about what can be known and how it should be justified.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

- Prepare intentionally awkward pairings, such as proving a poem or interpreting a chemical reaction as art.
- Frame the first round as playful so students can see method assumptions clearly.

Ready-to-Use Examples

- Question: Is this policy fair? Method card: mathematical proof.
- Question: What caused this revolution? Method card: controlled experiment.

Suggested Timing

Stage	Teacher move	Watch for
1	Groups draw one knowledge question and one method from different AOKs.	Assumptions, confidence shifts, evidence claims, and language choices.
2	They attempt to answer the question using the assigned method.	Assumptions, confidence shifts, evidence claims, and language choices.
3	Groups identify what the method reveals, distorts, or cannot handle.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Students swap to a more suitable method and compare results.	Assumptions, confidence shifts, evidence claims, and language choices.
5	Debrief why methods are powerful but not interchangeable.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

- A method is not just a tool; it carries a picture of what knowledge should look like.
- What did your method make impossible to see?

Common Misconceptions

- Students may apply one AOK's standard of evidence to all knowledge claims.
- Students may equate certainty with value or usefulness.

Assessment Look-Fors

- Compares standards of evidence across at least two AOKs.
- Explains how a claim changes when method, context, or audience changes.

Differentiation and Adaptations

- Support: provide obvious method matches after the mismatch round.
- Challenge: ask students to combine two methods and justify the combination.

Extension Tasks

- Students create their own method-swap card set for another class.
- Connect to interdisciplinary research and TOK essay comparisons.

Related Activities

- What Counts as Evidence?
- Same Object, Different AOK
- Reliability Ranking